DIY ELRS 4.0 Receiver — Wiring Guide

ESP32-C3 Zero + Ebyte E28-2G4M27SX (SX1281) — External Buck Converter Power

1. Power Setup (Buck Converter)

Power is supplied externally via a 3.3V buck converter. Both the ESP32-C3 Zero and the E28 module are powered from the same 3.3V buck output. The flight controller is NOT used as a power source.

⚠ NEVER connect 5V to ESP32-C3 Zero or E28 module. Both are 3.3V only. Reverse polarity or overvoltage will permanently damage them.

⚠ E28-2G4M27SX at full TX power can draw up to 700mA. Use a buck converter rated for at least 1A to be safe.

Component	Power Source	Voltage	Notes
Buck Converter Input	LiPo / Battery	7–26V typical	Depends on your buck module
Buck Converter Output	→ ESP32-C3 Zero 3.3V pin	3.3V	*Connect to 3.3V pin, NOT 5V*
E28 Module VCC	→ Same Buck 3.3V output	3.3V	*Parallel from same buck rail*
FC Power	Separate (FC's own BEC/ESC)	—	NOT connected to this circuit

ℹ Connect GND of buck, ESP32-C3 Zero, E28 module, and FC all together as a common ground reference even though FC power is separate.

2. SX1281 Radio Module Wiring (SPI)

Connect the E28-2G4M27SX module to the ESP32-C3 Zero using SPI. All pins are 3.3V logic.

E28 Module Pin	ESP32-C3 Zero GPIO	Wire Color (Suggested)	Notes
VCC	3.3V (Buck)	Red	*From buck output directly*

E28 Module Pin	ESP32-C3 Zero GPIO	Wire Color (Suggested)	Notes
GND	GND (Common)	Black	Buck, ESP32, module all share GND
SCK	GPIO6	Yellow	SPI Clock
MISO	GPIO5	Blue	SPI Data In
MOSI	GPIO4	Green	SPI Data Out
NSS / CS	GPIO7	Orange	Chip Select
BUSY	GPIO3	Purple	Radio Busy Signal
GDO2 (DIO1)	GPIO1	White	*IRQ Interrupt — moved off strapping pin*
NRESET	GPIO2	Grey	Safe for reset only
GDO0	Not connected	—	Leave floating, not needed by ELRS

⚠ GPIO2 is an ESP32-C3 strapping pin. It is safe to use for NRESET only because the reset line is driven after boot. Do NOT use GPIO2 for any other purpose.

ℹ GDO2 on the E28 module is the same as DIO1/IRQ on the bare SX1281. This is the interrupt line ELRS uses for packet detection.

3. CRSF Output to Flight Controller

Connect the receiver to your flight controller via UART. TX goes to RX and RX goes to TX — they always cross over.

ESP32-C3 Zero	Flight Controller Pad	Wire Color (Suggested)	Notes
GPIO21 (TX)	FC UART RX	Yellow	CRSF signal to FC
GPIO20 (RX)	FC UART TX	White	Telemetry from FC

⚠ Do NOT connect the FC's power/5V or 3.3V to this receiver. Power comes from the buck converter only.

> ℹ Only two wires are needed for CRSF — TX and RX. If you don't need telemetry, you can use TX only (one wire) but two wires is recommended for full ELRS telemetry.

4. Flight Controller Software Settings (Betaflight / INAV)

After wiring, configure your FC software to use CRSF on the UART you connected to:

Setting	Value
Serial RX Protocol	*CRSF*
UART Port	Whichever UART you connected GPIO20/21 to
Serial RX	Enabled on that UART
Telemetry	Enable if you want ELRS telemetry back

5. Pre-Power Checklist

1. Buck converter output measured at 3.3V before connecting anything
2. All GNDs connected together (buck, ESP32, E28, FC)
3. E28 VCC connected to 3.3V buck output (not FC)
4. SPI wiring: SCK→GPIO6, MISO→GPIO5, MOSI→GPIO4, NSS→GPIO7
5. BUSY→GPIO3, DIO1(GDO2)→GPIO1, RST(NRESET)→GPIO2
6. CRSF: ESP32 GPIO21(TX)→FC UART RX, GPIO20(RX)→FC UART TX
7. FC UART configured as Serial RX with CRSF protocol
8. ELRS firmware binding phrase matches your transmitter
9. Transmitter is on ELRS 4.0 (same major version required)